

Intellectual Merit Criterion

Overall Assessment of Intellectual Merit

Excellent

Explanation to Applicant

Two publications in review and several presentations as an undergraduate student are quite impressive. Nice job on winning the best undergrad talk award and receiving funding for those research efforts. This is what a potential NSF fellow should be embracing along the way. What I like the most about the applicant was drive and willingness to go above and beyond, volunteering to teach in Dev Biology courses, guest lectures, and mentoring underrepresented students in the lab. Beyond that, the applicant will assist with training and mentoring undergraduate students in the lab through Undergraduate Harlan Trust Research Program (credit-based undergraduate research) that may continue with independent research activities.

Broader Impacts Criterion

Overall Assessment of Broader Impacts

Excellent

Explanation to Applicant

I really liked applicants' BI - engagement in local School Without Walls program and Thomas Jefferson high school internship programs are great ideas and integrating these students not only lecturing them into the proposed studies to understand basic research is fantastic approach.

Summary Comments

The applicant presented a variety of undergraduate experiences, mentoring, teaching, publishing and above all, desire to make the difference in the world. The proposed project titled Amphibian immune-metabolic tradeoffs in the face of changing climates focuses on *Batrachochytrium dendrobatidis* (Bd) chytrid fungus that results in worldwide amphibian declines and extinctions. The proposed project has scientific merit and argument and hypotheses are clearly stated.

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Explanation to Applicant

The applicant has been involved in various research labs as an undergraduate student, involved disciplines ranging from recombinant gene expression in tobacco plants to investigating the impact of a sediment-dwelling worm on seagrass restoration. This dynamic range of experiences has allowed the applicant to fully appreciate the importance of cross-disciplinary research, as evidenced through the inception of an independent research project funded through a competitive undergraduate research award. The applicant continues to exhibit the capacity for independent thought based on the novel project proposal that has now developed into a doctoral project. This ability to integrate prior experiences together with the applicant's genuine curiosity for science will bode well for a career in research science. The applicant presents strong letters of recommendation that also highlight the applicant's scientific curiosity and resilience. This project proposes to investigate the consequences of environmental temperature-induced adaptation on amphibian metabolism and immune system performance. This proposal is driven by the overarching hypothesis that fungal infection driven increases in amphibian mortality reflect this immune-metabolic tradeoff. The applicant proposes to expose *X. laevis* froglets to cold, favorable, or hot conditions prior to Bd-challenge (both active Bd and heat-killed Bd). Skin samples will be harvested to assess immunological composition and metabolic states, thereby

establishing an understanding of climate change and amphibian immune health. The proposal contains a strong hypothesis, generated by the applicant, that follows a logical path of experiments. A few areas could, however, be improved in both clarity and conciseness. Based on Fig1, it appears that the applicant is separating innate (cold) versus adaptive (hot) immunity solely through temperature, when in fact, the applicant is assessing immune-metabolic readouts at cold versus hot. It is also unclear how gamma-irradiation will preferentially deplete lymphocytes, as it will likely target all hematopoietic-derived populations (e.g. myeloid). The proposal would also benefit with the inclusion of expected outcomes and identification of potential pitfalls (with alternative approaches). For example, what happens if temperature has no effect on immune health?

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Explanation to Applicant

The applicant has had several positive mentors and recognizes the importance of scientific mentoring and outreach. As such, the applicant has been involved in several volunteer and teaching activities at all levels. The applicant also details future plans to increase heterogeneity in academia and to educate the community. This includes engaging underrepresented groups in academia through a matching program, and the development of social media platforms to promote awareness of ecology-related concerns. The applicant is highly encouraged to pursue these planned activities.

Summary Comments

Overall the applicant is well-recommended and demonstrates the maturity for a successful researcher. The applicant has been involved in several research labs, where each experience has led to several conference presentations and pending publications. While positive mentoring is likely involved, this is highly reflective of the applicant's ability to complete projects in a timely fashion. The applicant's proposal clearly states an unknown question with relevant aims, where further research will benefit our understanding of climate change and amphibian immune adaptation. While the research proposal would benefit from additional clarity, this will likely come with experience. Overall, the application is strong and I believe the applicant will be a successful graduate student and researcher.

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Overall Assessment of Intellectual Merit

Very Good

Explanation to Applicant

I found the figure helpful and a nice inclusion. Some definitions could be given: trade-offs for example.

Broader Impacts Criterion

Overall Assessment of Broader Impacts

Very Good

Explanation to Applicant

I really liked the conservation aspect (talks on amphibian declines to the general public). Some details on how public awareness will be increased would be helpful.

Summary Comments

An interesting project that may benefit the conservation of amphibians through the use of cutting-edge methodologies.

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